

A CLASS OF EQUIVALENT PROBLEMS IN STATISTICS AND OPERATIONAL RESEARCH

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Summary: A broad class of equivalent problems in Statistics and Operational Research, including problems relating to nonparametric maximum likelihood estimation for finite mixture models, to indirect estimation involving Poisson processes, to the construction of Bayesian D-optimal designs for one-parameter nonlinear models, and to the Kalai-Nash solution to simple bargaining games, is discussed. In particular some key results pertaining to the generic problem are drawn together from the finite mixture distribution and the optimal design literature and the advantages which accrue from recognizing this class of equivalent problems are emphasized. The import of the findings is highlighted by means of examples involving maximum likelihood image reconstruction for PET scans and the selection of optimal investment portfolios.

1. Introduction

There is a broad class of optimization problems in a diversity of areas in Statistics and Operational Research which have the following form.

Find the probability measure, $F(\cdot)$, defined on the space, $\mathcal{X}$, which

maximizes

$$\Phi(F) = \sum_{j=1}^p g_j \ln \left[\int_{x \in \mathcal{X}} h_j(x) dF(x) \right] \quad (1)$$

where $h_j(\cdot)$ is a specified function on $\mathcal{X}$, g_j is given, and $dF(\cdot)$ represents the Stieljes integral.

This class of problems has been drawn together within the framework of the EM algorithm by Vardi and Lee (1993), and their notation is adopted here. The aim of the present study is to pursue the remarks of Lindsay (1993) in the discussion of the paper by Vardi and Lee and those of Haines (1995, 1996), and to demonstrate that it is the *equivalence* of the problems themselves, rather than the *algorithm* used to solve them, which is of key importance. More specifically, the intention here is to identify and to explore this class of problems, and to highlight the advantages that accrue from formalizing their equivalence.

This article is expository in nature and aims to provide the practitioner who has identified a problem mapping onto Problem (1) with a suite of appropriate and powerful results. The generic problem is considered in detail in Section 2, and in particular key results based on convex geometry, some powerful theorems, and algorithms providing numerical solutions to the problem are presented. In Section 3 a selection of problems, which pertain to inference and to decision theory and which are equivalent to Problem (1), are discussed, compared and contrasted. Some broad conclusions and pointers for future research are given in Section 4.

2. The generic problem

The results, theorems and algorithms presented here for Problem (1) are embedded in the theory of optimal experimental design (Fedorov, 1972; Silvey, 1980; Haines, 1995) and are particularly clearly and concisely explained within the context of nonparametric maximum likelihood estimation for finite mixture distributions by Lindsay (1983 a,b; 1995).

2.1 Theory

The function values, $h_j(x)$, $j = 1, \dots, p$, $x \in \mathcal{X}$, introduced in the formulation of Problem (1), can be assembled as the set of p -tuples,

$$C = \{(h_1(x), \dots, h_p(x)) \mid x \in \mathcal{X}\},$$

and the convex hull of this set, which is the set of all such p -tuples taken over the class of probability measures on $\mathcal{X}$ with finite support, can be expressed as

$$S = \text{conv}(C) = \left\{ \left(\sum_{i=1}^k f_i h_1(x_i), \dots, \sum_{i=1}^k f_i h_p(x_i) \right) \mid x_i \in \mathcal{X}, f_i > 0, \sum_{i=1}^k f_i = 1 \right\}.$$

Note that the set, C , either defines a parametric curve or surface in $\mathbb{R}^p$, or, for $\mathcal{X}$ finite, comprises isolated points in $\mathbb{R}^p$. Furthermore, if the set, C , is compact, then S is also compact, and the set,

$$S^* = \left\{ \left(\int_{x \in \mathcal{X}} h_1(x) dF(x), \dots, \int_{x \in \mathcal{X}} h_p(x) dF(x) \right) \mid F \in \mathcal{F} \right\},$$

where $\mathcal{F}$ denotes the class of *all* probability measures on $\mathcal{X}$, coincides with S (Lindsay, 1983a, p. 87). The following key result can now be readily deduced from the concavity of the function, $\Phi(F)$, on the class of probability

measures, $\mathcal{F}$, from the definition of the set, S^* , and from Carathéodory's Theorem (Fedorov, 1972, pp. 66, 79; Silvey, 1980, pp. 16–17; Lindsay, 1983a).

Theorem 1. If the set, C , is compact, then the function, $\Phi(F)$, has a unique maximum and furthermore there always exists a discrete probability measure, F^* , corresponding to that maximum, which is based on at most p points of support.

The directional derivative of the function, $\Phi(F)$, at a point, $x \in \mathcal{X}$, in the direction of a probability measure, F , is taken to be the Fréchet derivative,

$$\lim_{\epsilon \rightarrow 0^+} \frac{\Phi[(1 - \epsilon)F + \epsilon \delta_x] - \Phi(F)}{\epsilon},$$

where δ_x represents the measure which puts mass 1 at the point x (Silvey, 1980, p. 18). In the present case this derivative is readily shown to have the form

$$D(x, F) = \sum_{j=1}^p g_j \left[\frac{h_j(x)}{\int_{x \in \mathcal{X}} h_j(x) dF(x)} - 1 \right],$$

and an Equivalence Theorem, relating optimization problems for $\Phi(F)$ and for $D(x, F)$, and based on the celebrated theorem of Kiefer and Wolfowitz (1960), can be formulated as follows (Lindsay, 1983a, Theorem 4.1; Chaloner and Larntz, 1989).

Theorem 2. A probability measure, F^* , can be equivalently characterized by any one of the three conditions.

1. F^* maximizes $\sum_{j=1}^p g_j \ln \left[\int_{x \in \mathcal{X}} h_j(x) dF(x) \right]$,
2. F^* minimizes $\sup_{x \in \mathcal{X}} D(x, F)$

$$3. \sup_{x \in \mathcal{X}} D(x, F^*) = 0.$$

Furthermore, the support of F^* is contained in the set of values of $x \in \mathcal{X}$ for which $D(x, F^*) = 0$.

This theorem is *remarkably* powerful from a practical point of view, since it provides a mechanism for confirming the global optimality or otherwise of a solution to Problem (1), which is a problem in nonlinear optimization, by means of an examination of the directional derivatives. It should be noted however that the probability measure, F^* , characterized in Theorem 2, and maximizing $\Phi(F)$, is not necessarily unique. A sufficient condition for its uniqueness is that the set of vectors,

$$\{(h_1(x), \dots, h_p(x)) \mid x \in \mathcal{X}, D(x, F^*) = 0\},$$

are linearly independent (Lindsay, 1983a; Mallet, 1986), but this condition is not always easy to verify in practice since *all* the values of $x \in \mathcal{X}$ satisfying $D(x, F^*) = 0$ must be identified.

There are many other interesting results relating to Problem (1). For example a dual to this problem follows from the notion of the directional derivative for the concave function, $\Phi(F)$, (Lindsay, 1983a, Theorem 5.1), and can be expressed as

$$\text{maximize } \sum_{j=1}^p g_j \ln w_j \quad \text{subject to} \quad \sum_{j=1}^p w_j h_j(x) \leq \sum_{j=1}^p g_j \quad \text{for all } x \in \mathcal{X},$$

for variables, $w_j > 0, j = 1, \dots, p$. This duality result provides additional geometric insight into the nature of Problem (1), but is not of great practical value. Further, for the case where $p = 2$, the convex geometry described

above can be represented graphically and this representation can in turn be used to obtain explicit expressions for the solutions to Problem (1) and to provide valuable insights into theoretical results such as Carathéodory's Theorem and the Equivalence Theorem (Lindsay, 1983a,b; Haines, 1995). Such an approach is however clearly limited to two, or at most three, dimensions.

2.2 Algorithms

Solutions to Problem (1) are in general obtained numerically, and it is clear from the theory described above that the search for such solutions can be restricted to the class of *discrete* probability measures on $\mathcal{X}$, that an optimal solution based on at most p points of support exists, and that the global optimality or otherwise of a candidate solution can be confirmed by invoking the Equivalence Theorem.

A widely-used strategy for finding a solution to Problem (1), termed the vertex direction method (Fedorov, 1972; Wu, 1978) and based on the Equivalence Theorem, proceeds as follows.

Algorithm 1.

Step 1. Select F_1 , a probability measure on $\mathcal{X}$. Set $m = 1$.

Step 2. Find $x_m^* = \arg \max_{x \in \mathcal{X}} D(x, F_m)$. If $D(x_m^*, F_m) \leq 0$, stop.

Step 3. Find the step-length,

$$\epsilon_m^* = \arg \max_{\epsilon \in [0,1]} \Phi[(1 - \epsilon)F_m + \epsilon \delta_{x_m^*}],$$

and take

$$F_{m+1} = (1 - \epsilon_m^*)F_m + \epsilon_m^* \delta_{x_m^*}.$$

Set $m = m + 1$, and go to Step 2.

It is immediately clear that large numbers of distinct points can accumulate during this iterative procedure. This problem can be alleviated either by modifying Step 3 so that the probability measure, F_{m+1} , calculated at the m th iteration, has points of support taken from those of F_m together with the point, x_m^* , and weights assigned to the selected points in such a way that $\Phi(F_{m+1})$ is maximized (Wu, 1978), or by various other methods (see e.g. Fedorov, 1972, p. 109; Mallet, 1986).

Several extensions to Algorithm 1, aimed at reducing the number of iterations required for convergence, are available. Of these, the vertex exchange method, which, at the m th iteration, transfers some or all of the mass from the support point of F_m minimizing the directional derivative, $D(x, F_m)$, to a point in $\mathcal{X}$ maximizing that derivative, is perhaps the most widely used (Böhning, 1985, 1986). A more recent approach, termed the intra-simplex direction method, involves augmenting the probability measure at the m th iteration, F_m , with *all* points, $x \in \mathcal{X}$, maximizing $D(x, F_m)$ such that $D(x, F_m) > 0$, and distributing mass optimally between F_m and these points (Lesperance and Kalbfleisch, 1992). Proofs of convergence for these vertex direction and exchange algorithms are based on that given by Fedorov (1972, pp. 102–103) for D-optimal designs.

The EM algorithm is also used extensively for finding solutions to Problem (1), and can be formulated succinctly as follows.

Algorithm 2.

Step 1. Select F_1 , a discrete probability measure on $\mathcal{X}$ with r support points, $x_i, i = 1, \dots, r$. Set $m = 1$

Step 2. Find the probabilities,

$$f_i^{(m+1)} = \frac{f_i^{(m)}}{\sum_{j=1}^p g_j} \sum_{j=1}^p \frac{g_j h_j(x_i)}{\sum_{k=1}^r f_k^{(m)} h_j(x_k)},$$

for $i = 1, \dots, r$.

Step 3. If the convergence criterion is met, stop. Otherwise set $m = m + 1$, more over also and go to Step 2.

The procedure requires that a fixed number of support points, r , be used, but it follows immediately from Theorem 1 that this is not restrictive. In particular if the space, $\mathcal{X}$, comprises a finite set of support points, it is usual to retain all such points in the algorithm and, at convergence, to equate an appropriate number of suitably small probabilities to zero. This approach is the one most frequently used in applications. Alternatively, if the space, $\mathcal{X}$, is assumed to be infinite, it is sensible to take the number of support points to be equal to p , and, following Redner and Walker (1984, p. 217), to modify Step 2 to include the updating of the values of the support points according to the scheme,

$$f_i^{(m+1)} = \frac{f_i^{(m)}}{\sum_{j=1}^p g_j} \sum_{j=1}^p \frac{g_j h_j(x_i^{(m)})}{\sum_{k=1}^r f_k^{(m)} h_j(x_k^{(m)})},$$

and

$$x_i^{(m+1)} = \arg \max_{x \in \mathcal{X}} \sum_{j=1}^p \frac{g_j h_j(x_i^{(m)})}{\sum_{k=1}^r f_k^{(m)} h_j(x_k^{(m)})} \ln h_j(x).$$

The formula for updating the probabilities given in Step 2 of this EM algorithm emerges naturally from the idea of maximizing the log function, $\Phi(F)$, by using multiplicative steps involving its gradient (Silvey, Titterington,

and Torsney, 1978; Cover, 1984). More structured derivations of the complete algorithm are also available and, in the spirit of the EM algorithm, are based on settings with missing observations (Redner and Walker, 1984; Vardi, Shepp and Kaufman, 1985; Vardi and Lee, 1993). Proof of convergence of Algorithm 2 follows immediately from the corresponding proof for the general EM algorithm (Redner and Walker, 1984). However convergence to a global rather than a local optimum is *not* guaranteed and in this context the value of invoking Theorem 2 in order to confirm the global optimality or otherwise of a candidate solution cannot be overemphasized.

Algorithm 2 enjoys all the attractive features of an EM algorithm. In particular, the required constraints, $f_i \geq 0, i = 1, \dots, r$, and $\sum_{i=1}^r f_i = 1$, are automatically conserved at each iteration, convergence is monotonic and, in general, well-behaved, and computation is necessarily efficient (see e.g. Redner and Walker, 1984). Naturally, the algorithm also suffers from the disadvantage common to all EM algorithms, in that convergence, after the first few iterations, is notoriously slow. Methods for accelerating this rate of convergence have been devised (Silvey, Titterton, and Torsney, 1978; Torsney, 1993), but these tend to detract from the simplicity and elegance of the basic EM algorithm.

Problem (1) is a problem in constrained optimization, and standard techniques, such as projection methods based conjugate gradient procedures (Wu, 1978), the use of appropriate penalty functions together with the simplex algorithm (Chaloner and Larntz, 1989), and the conversion of the problem to an unconstrained one by means of an appropriate trigonometric transformation of the weights (Atkinson, 1975), can be used to solve it. Many of these methods

do however require good starting values, and in addition convergence is not always stable.

3. Examples

3.1 Estimation

Finite Mixture Distributions

Suppose that $f(\tau, y)$ represents a family of probability density functions relating to a random variable, Y , and indexed by a parameter, τ , from the parameter space, T , and suppose further that the distribution of the random variable, Y , is described by the mixture density,

$$f_Q(y) = \int_{\tau \in T} f(\tau, y) dQ(\tau),$$

where Q is a probability measure on T and is termed a mixing distribution. Then, for a random sample, $y_1, \dots, y_p$, taken from this mixture density, the likelihood function is given by $\prod_{j=1}^p [f_Q(y_j)]$ and the nonparametric maximum likelihood estimator (NPMLE) for the mixing distribution maximizes this function, or equivalently its logarithm,

$$\sum_{j=1}^p \ln \int_{\tau \in T} f(\tau, y_j) dQ(\tau).$$

If the random variable, Y , is discrete, this log-likelihood can be written more compactly in terms of *distinct* values, y_j , replicated m_j times for $j = 1, \dots, p$ respectively, as

$$\sum_{j=1}^p m_j \ln \int_{\tau \in T} f(\tau, y_j) dQ(\tau).$$

This problem of maximum likelihood estimation is precisely the generic Problem (1).

The results described in Section 2 emanate from the literature on this example, and have indeed been widely applied to finite mixtures of both continuous and discrete distributions (Lindsay, 1995). In addition likelihood-based results on inference for mixing distributions are available (Lindsay, 1995), but are not relevant in the context of decision theory and are therefore not discussed here. Overall it is clear that the problem of obtaining NPMLE for finite mixture distributions must be regarded as the prototype for Problem (1).

Indirect Estimation involving Poisson Distributions

Suppose that interest focuses on estimating the intensity function, $\lambda(x)$, of a nonhomogeneous Poisson process defined on a space, $\mathcal{X}$. Suppose however that events from this process are not directly observable but that events from the related nonhomogeneous Poisson process defined on the space, $\mathcal{Z}$, and having intensity function

$$g(z) = \int_{\mathcal{X}} K(x, z) \lambda(x) dx,$$

with $K(\cdot, \cdot)$ a kernel function proportional to a conditional density, are observable. Then the intensity function, $\lambda(x)$, can be estimated from indirect measurements made on this latter process using maximum likelihood. Specifically, suppose that the data are available in the form of independent counts, $n_j, j = 1, \dots, p$, for p nonoverlapping bins which exhaustively cover the space $\mathcal{Z}$, and suppose further that the space $\mathcal{X}$ is similarly discretized into k bins. Then the counts, n_j , are independent Poisson variates with means, $\sum_{i=1}^k \lambda_i p_{ij}$, where λ_i is the intensity function for the i th bin of $\mathcal{X}$ and p_{ij}

is the probability that an event occurring in the i th bin of $\mathcal{X}$ is recorded in the j th bin of $\mathcal{Z}$ for $i = 1, \dots, k$ and $j = 1, \dots, p$ respectively. The log-likelihood function is thus given by

$$\sum_{j=1}^p n_j \ln \left(\sum_{i=1}^k \lambda_i p_{ij} \right) - \sum_{j=1}^p \left(\sum_{i=1}^k \lambda_i p_{ij} \right) - \sum_{j=1}^p \ln(n_j!), \quad (2)$$

and hence, provided the probabilities p_{ij} are scaled so that $\sum_{j=1}^p p_{ij} = 1$, with the intensities λ_i scaled accordingly, for $i = 1, \dots, k$ and $j = 1, \dots, p$, by

$$\sum_{j=1}^p n_j \ln \left(\sum_{i=1}^k \lambda_i p_{ij} \right) - \sum_{i=1}^k \lambda_i - \sum_{j=1}^p \ln(n_j!).$$

It then follows from the fact that the likelihood is concave and from the Kuhn-Tucker conditions that the problem of maximizing this log-likelihood is equivalent to that of maximizing

$$\sum_{j=1}^p n_j \ln \left(\sum_{i=1}^k \lambda_i p_{ij} \right) \quad \text{subject to} \quad \sum_{i=1}^k \lambda_i = \sum_{j=1}^p n_j = n,$$

and hence, by scaling the estimated emission densities with the constant n , to Problem (1). It should be noted that the effect of the constraint $\sum_{j=1}^p p_{ij} = 1$ for $i = 1, \dots, k$ is simply to reduce the dimensionality of the overall problem by 1. The salient features of the mapping of this problem onto that of obtaining NPMLE for finite mixture distributions are summarized in Table 1.

As an aside it is interesting to observe that the Poisson model described by the log-likelihood (2) is in fact a generalized linear model with Poisson error, an identity link function, and parameters constrained to be nonnegative (Green, 1990) and that consequently all the results of the present study also hold for such generalized linear models. In addition it is interesting to note

that Lindsay (1995, p. 10) formulated the problem of intensity estimation from indirect measurements as a problem of obtaining the NPMLE of a mixture density and in so doing obviated the need to assume a Poisson process.

The approach to maximum likelihood estimation for the intensity function of a Poisson process from indirect measurements described above is widely used. Examples include image reconstruction of positron emission tomography (PET) scans (e.g. Shepp and Vardi, 1982; Vardi, Shepp and Kaufman, 1985; Vardi and Lee, 1993), single-photon emission computerized tomography (Green, 1990), a classical problem in stereology (Silverman, Jones, Wilson, and Nychka, 1990), and the estimation of particle size distributions from diffusion battery data (Maher and Laird, 1985; Laird and Maher, 1985). In all cases studied interest has focussed almost exclusively on the use of the EM algorithm, and variants of it, to obtain the estimated intensities. It is now clear that the broader theory and results of Section 2 can be applied to these examples and in addition that a wide range of algorithms for obtaining the required estimates can be meaningfully explored.

Consider for example the PET scan scenario in which pixels comprising the image of a patient's brain correspond to bins in the space, $\mathcal{X}$, and detector tubes defined by the detector units in a band circling the patient's head correspond to bins in the space, $\mathcal{Z}$. Many researchers have reported that maximum likelihood reconstructions of these scans tend to produce images in which an unduly large number of pixels have zero or near-zero estimated emission densities and which therefore exhibit 'checkerboard' patterns (e.g. Vardi, Shepp and Kaufman, 1985; Green, 1990; Silverman, Jones, Wilson, and Nychka, 1990). This effect can be readily explained by observing that in a typical PET scan

the number of pixels is very much greater than the number of detector tubes, i.e. $k \gg p$, but that by Theorem 1 there always exists an image in which the number of pixels with non-zero estimated emission densities is less than or equal to the number of detector tubes. Such images will therefore exhibit 'checkerboard patterns' and these images, or ones similar to them, are clearly realized in practice.

Other insights into the problem of reconstructing PET scan images by using maximum likelihood also emanate from its equivalence with Problem (1). In particular Theorem 2 can be invoked to confirm the global optimality or otherwise of a reconstructed image. Furthermore algorithms for solving Problem (1) discussed in Section 2, other than the EM algorithm, can also be invoked in the PET scan setting with the proviso that for large numbers of pixels and of detector tubes the necessary matrix manipulations may be prohibitive in terms of storage requirements (Hudson, Ma, and Green, 1994).

3.2 Decision Theory

Bayesian D-Optimal Designs for One-parameter Nonlinear Models

Suppose that a nonlinear model involves a single parameter, θ , and a single explanatory variable, x , where x belongs to a design space, $\mathcal{X}$, and suppose further that the information for θ at a design point, $x \in \mathcal{X}$, is given by $I(x, \theta)$. Then the information for a design measure, μ , on $\mathcal{X}$, can be expressed as

$$I(\mu, \theta) = \int_{\mathcal{X}} I(x, \theta) d\mu(x),$$

and it is clearly desirable and meaningful to construct and use designs, μ , which maximize the information, $I(\mu, \theta)$, or some function thereof. Since

$I(\mu, \theta)$ depends on the unknown parameter, θ , this is not a straightforward matter. One approach to the construction of optimal designs in such cases is to express the experimenter's prior belief in the parameter space by adopting a discrete prior distribution, β , which puts weights, β_j , on the parameter values, $\theta_j, j = 1, \dots, p$, respectively, and to maximize the function,

$$\sum_{j=1}^p \beta_j \ln \left[\int_{\mathcal{X}} I(x, \theta_j) d\mu(x) \right]$$

with respect to the design measure, μ . Designs maximizing this criterion are examples of Bayesian D-optimal designs (Chaloner and Larntz, 1989), and the problem of their construction is clearly equivalent to Problem (1). The nature of this equivalence is highlighted in Table 1.

Many of the results presented in Section 2 emanate from the optimal design literature, and their application to Bayesian D-optimal designs for one-parameter nonlinear models is therefore well known (Chaloner, 1993; Haines, 1995; Dette and Neugebauer, 1996). There are however further insights into the construction of such designs which can be gleaned from the cohesive theory of Section 2, and in addition powerful results can be borrowed from the literature on NPMLE for finite mixture distributions. Thus, for example, Lindsay (1983a) has developed some interesting results pertaining to mixing distributions with a restricted number of support points which find immediate application in the optimal design setting.

Simple Bargaining Games

Suppose that a decision maker, B , associates a utility, $U(a, B)$, with an action, a , from some action space, $\mathcal{A}$, and suppose further that the decision maker is

prepared to accept a randomized decision rule, δ , on $\mathcal{A}$, and to associate a utility, $\int_{\mathcal{A}} U(a, B) d\delta(a)$, with that rule. Consider a simple bargaining game involving p such decision makers. Then the Kalai-Nash solution to this bargaining game is obtained by maximizing the product $\prod_{j=1}^p [\int_{\mathcal{A}} U(a, B_j) d\delta(a)]^{\alpha_j}$ or equivalently its logarithm

$$\sum_{j=1}^p \alpha_j \ln \left[\int_{\mathcal{A}} U(a, B_j) d\delta(a) \right],$$

with respect to the randomized rule δ , where the terms, α_j , represent weights given to the opinions of the decision makers, B_j $j = 1, \dots, p$ respectively (Nash, 1950; Kalai, 1977). The problem of obtaining this solution is clearly equivalent to Problem (1) and hence to that of obtaining the NPMLE of a mixing distribution, to that of indirect estimation for Poisson processes, and to that of constructing Bayesian D -optimal designs for one-parameter non-linear models. Details of these latter equivalences are given in Table 1. In fact it is interesting to observe here that the formulation of Bayesian D -optimal designs can be regarded as a simple bargaining game and can thus be considered to be subsumed within that paradigm.

Examples of simple bargaining games usually involve two decision makers and include cooperative bimatrix games, for which the action space $\mathcal{A}$ is finite (e.g. Owen, 1982, p. 140), and games involving the sharing of a sum of money between two players, for which $\mathcal{A}$ is infinite (e.g. Owen, 1982, p. 136). Further examples arise in a statistical context from the work of Weerahandi and Zidek (1981) in the area of multi-Bayesian decision theory. In general however for bargaining games involving more than two players it is usual to abandon the Kalai-Nash solution, and to invoke bargaining models based on coalitions. One

exception to this occurs in the area of optimal investment portfolios. In this case the actions correspond to stocks T_i $i = 1, \dots, k$, the decision rule δ represents a portfolio of these stocks, and the utility of the stock T_i for the decision maker B_j is the annual return on \$1, R_{ij} , that the decision maker believes the stock will yield where $i = 1, \dots, k$ and $j = 1, \dots, p$. For an investor whose confidence in the decision maker B_j is expressed as the probability α_j , where $j = 1, \dots, p$ and $\sum_{j=1}^p \alpha_j = 1$, the problem of finding an optimal investment portfolio now becomes that of maximizing

$$\sum_{j=1}^p \alpha_j \ln \left[\sum_{i=1}^k \delta_i R_{ij} \right].$$

It is both surprising and reassuring to observe that this Kalai-Nash solution coincides with the solution derived by Cover (1984) from arguments based on expected log annual returns.

Work on the Kalai-Nash solution to simple bargaining games has focussed almost exclusively on characterizing optimal decision rules for two decision makers graphically (e.g. Weerahandi and Zidek, 1983; de Waal, Groenewald, van Zyl, and Zidek, 1986) and, in the case of optimal investment portfolios, on devising an algorithm, which is equivalent to the EM algorithm, for finding the optimal portfolio (Cover, 1984; Vardi and Lee, 1993). The broad theory, results and algorithms relating to Problem (1) and outlined in Section 2 therefore find immediate and powerful application within the context of simple bargaining games, and in particular within the area of optimal investment portfolios. Thus for example it follows from Theorem 1 that an optimal investment portfolio in which the number of stocks is less than or equal to the number of decision makers consulted always exists, an observation which is clearly of some

practical interest. Furthermore optimal portfolios can be constructed using, for example, vertex exchange methods and their global optimality confirmed by invoking the Equivalence Theorem.

4. Conclusions

The main purpose of this paper has been to demonstrate that there are many problems in Statistics and Operational Research which are equivalent to each other and in particular to the prototype problem in inference of obtaining NPMLE for finite mixture distributions and in decision theory of constructing Bayesian D-optimal designs for one-parameter nonlinear models. A selection of these problems has been introduced here and others can be identified from the work of Vardi and Lee (1993) and include solving integral equations by minimizing a Kullback-Leibler distance, obtaining maximum likelihood estimates for grouped data, and estimating traffic intensities in network tomography (Vardi, 1996).

An immediate and fascinating point of interest is the fact that problems in inference and in decision theory are linked through the class of equivalent problems characterized by (1). Furthermore the key features of this class of problems which are developed in this paper and which are of particular relevance to the practitioner are that optimal solutions to the problems based on at most p points of support always exist, that a wide range of algorithms for finding solutions is available, and that an Equivalence Theorem can be invoked to confirm the global optimality or otherwise of a candidate solution. These results can be used to some advantage, as is demonstrated in the present study for the problem of image reconstruction for PET scans and of specifying an

optimal investment portfolio. Detailed accounts of these applications, including some more technical aspects, are available in separate papers elsewhere.

Two extensions to the present study follow immediately and both of these provide scope for future research. Firstly it is clear that the results of Section 2 hold for any problem of the form (1) which involves the maximization of a sum,

$$\sum_{j=1}^p g_j \Psi \left[\int_{x \in \mathcal{X}} h_j(x) dF(x) \right],$$

where the function, Ψ , is concave. It would thus undoubtedly be of some interest to identify broad classes of problems which can be formulated in this manner. Secondly it would seem natural to consider the problem of maximizing the integral

$$\int_{y \in \mathcal{Y}} \ln \left[\int_{x \in \mathcal{X}} h(x, y) dF(x) \right] dG(y), \quad (3)$$

with respect to the probability measure, F , where the space, $\mathcal{Y}$, is taken to be infinite and $h(\cdot, \cdot)$ is a specified function on $\mathcal{X} \times \mathcal{Y}$. This problem is of some relevance, as for example in the construction of Bayesian D-optimal designs with a continuous prior distribution. Vardi and Lee (1993) have developed an EM algorithm for finding solutions to the maximization of the integral, (3), but the numerical implementation of their procedure would seem, at least in general, to require discretization of the space, $\mathcal{Y}$. Furthermore it is clear that much of the theory of Section 2 will break down in this more general case and indeed L  uter (1974) has demonstrated that the upper bound on the number of support points provided by Carath  odory's Theorem no longer holds. Overall therefore the problem of maximizing the integral (3) is a particularly challenging one, both analytically and numerically.

5. References

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Table 1. A comparison of equivalent problems.

NPML estimation for finite mixture distributions	Indirect estimation for Poisson processes	Bayesian D-optimal designs	Simple bargaining games
P.d.f. $f(\tau, y)$	Probability p_{ij}	Information $I(x, \theta)$	Utility $U(a, B)$
Mixing distribution $Q = \left\{ \begin{matrix} \tau_1, \dots, \tau_k \\ q_1, \dots, q_k \end{matrix} \right\}$ with $f_Q(y) = \sum_{i=1}^k q_i f(\tau_i, y)$	Intensities $\lambda_1, \dots, \lambda_k$ on k pixels with $\eta_j = \sum_{i=1}^k \lambda_i p_{ij}$	Approximate design $\mu = \left\{ \begin{matrix} x_1, \dots, x_k \\ \mu_1, \dots, \mu_k \end{matrix} \right\}$ with $I(\mu, \theta) = \sum_{i=1}^k \mu_i I(x_i, \theta)$	Randomized rule $\delta = \left\{ \begin{matrix} a_1, \dots, a_k \\ \delta_1, \dots, \delta_k \end{matrix} \right\}$ with $U(\delta, B) = \sum_{i=1}^k \delta_i U(a_i, B)$
Random sample $\left\{ \begin{matrix} y_1, \dots, y_p \\ m_1, \dots, m_p \end{matrix} \right\}$	Counts $n_1, \dots, n_p$ in p detector tubes	Parameters $\left\{ \begin{matrix} \theta_1, \dots, \theta_p \\ \beta_1, \dots, \beta_p \end{matrix} \right\}$	Decision makers $\left\{ \begin{matrix} B_1, \dots, B_p \\ \alpha_1, \dots, \alpha_p \end{matrix} \right\}$
NPML estimation maximize $\sum_{j=1}^p m_j \ln \left\{ \sum_{i=1}^k q_i f(\tau_i, y_j) \right\}$	Likelihood criterion maximize $\sum_{j=1}^p n_j \ln \left\{ \sum_{i=1}^k \lambda_i p_{ij} \right\}$	Bayesian D-optimality maximize $\sum_{j=1}^p \beta_j \ln \left\{ \sum_{i=1}^k \mu_i I(x_i, \theta_j) \right\}$	Kalai-Nash solution maximize $\sum_{j=1}^p \alpha_j \ln \left\{ \sum_{i=1}^k \delta_i U(a_i, B_j) \right\}$